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DP2Net: A discontinuous physical property-constrained single-source domain generalization network for tool wear state recognition

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ABSTRACT

Cross-conditions tool wear monitoring has a wide application prospect in manufacturing. However, the data distribution discrepancies caused by the inconsistency of process elements restrict the generalization of models under cross-conditions or even similar conditions. The existing methods based on diversity enhancement make it difficult to effectively establish the correlation between the source domain and target domains, which limits the improvement of model generalization. Therefore, this paper details the cause of data distribution discrepancies and proposes a discontinuous physical property-constrained single-source domain generalization network for milling tool wear monitoring. Firstly, a spatial attention mechanism is introduced to weight key signal segments adaptively. Secondly, the generation module is constrained by a standard sample and is used to generate diverse samples with physical properties. Thirdly, extensive recognition experiments on an open dataset and machining experiments with three distribution discrepancy levels were conducted to verify the effectiveness of the proposed method. Finally, feature visualizations provide consistency and interpretability.

1. Introduction

Tools are the key executors of machine tools, known as the teeth of the manufacturing industry. They are always in a continuous wear state, and their excessive wear will deteriorate product quality and equipment performance, increase manufacturing costs, and even cause unsafe accidents [1]. Real-time and precise tool wear monitoring has always been crucial in modern manufacturing.

Industrial cameras and other direct measuring techniques accurately assess tool wear under any working conditions, but the machining must be paused, significantly reducing production efficiency. Experiments under specific conditions determine the key parameters of traditional physical [2] or statistical models, so the generalization is challenging to meet the practical application. At

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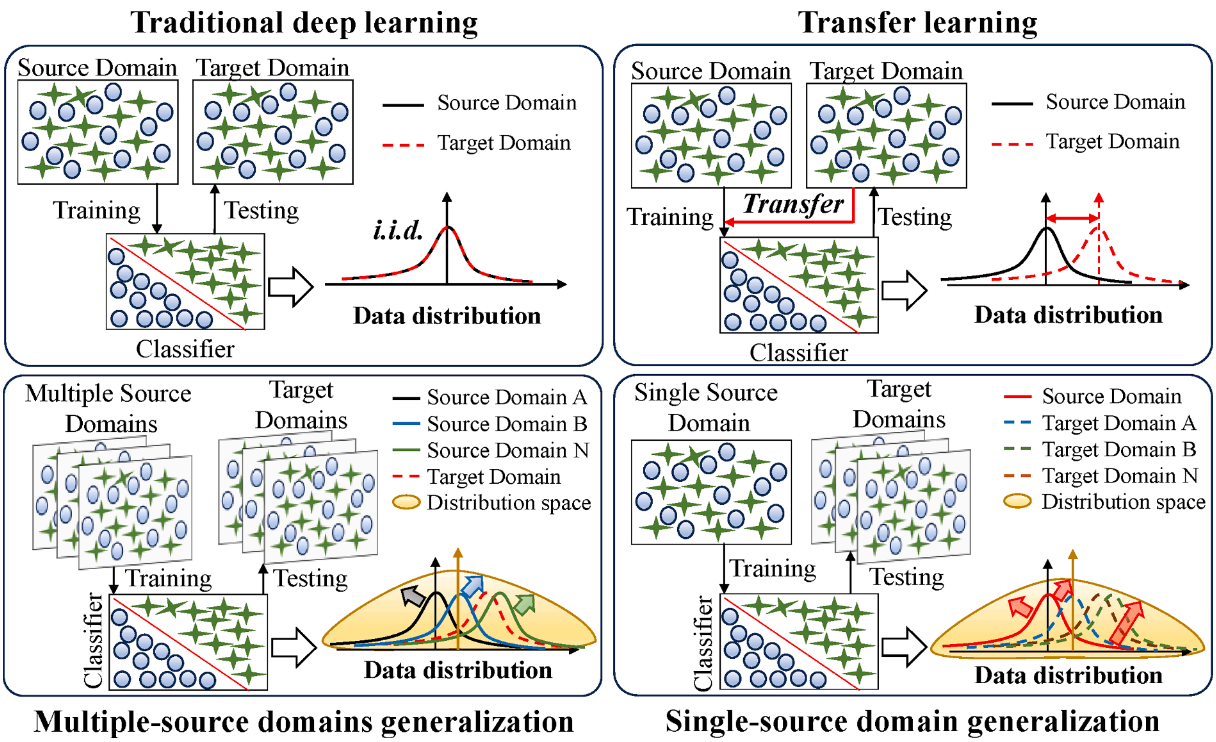

Fig. 1. Illustration of traditional DL, TL, MSDG, and SSDG.

present, the data-driven approach represented by deep learning (DL) has attracted the common attention of academia and industry [34]. DL aims to establish a nonlinear mapping relationship between the physical signal and the tool wear in a high-dimensional space. Xue *et al.* [5] carried out interactive weighting of the features of different channel monitoring signals with the channel attention mechanism and convolutional neural network, effectively improving the accuracy of tool wear state recognition. Liu *et al.* [6] proposed a multi-input parallel convolution structure for multi-scale feature extraction of signals from different sensors during milling and adaptive differentiation of the importance of different signals. Guo *et al.* [7] explored a mechanism-guided interpretable tool wear monitoring DL model. Zhou *et al.* [8] conducted a study on small sample scenarios in tool wear monitoring. Zou *et al.* [9] discussed sample imbalance in tool wear monitoring. Qiang *et al.* [10] improved the generalization of the tool wear monitoring model based on transfer learning (TL) with limited target domain data. Leon *et al.* [11] provide a comprehensive review of relevant methods.

With the development of manufacturing technology and the change in production mode, the structure and material of tools and the machining conditions are increasingly complicated and varied, which undoubtedly challenge the generalization of tool wear state monitoring methods. Traditional DL relies on the assumption of independent and identical distribution (i.i.d.). That means there are no data distribution discrepancies (DDD) between the historical data used for model training and the data to be inferred. This assumption is not consistent with the actual situation. As shown in Fig. 1, TL imparts knowledge, such as target domain data/features, into the model in the training process, allowing the model parameters to update under pre-adaptation to the data distribution to be inferred [12]. TL greatly improves the generalization performance of the model on target domains. However, it is uncertain whether the TL model can be used for data outside of the predetermined target domains [13]. At the same time, a basic demand for a tool wear monitoring model is that the model trained on the data of the tool-cutting processes can be applied to other tools of the same type under exactly the same or slightly different conditions. TL methods have difficulty meeting the actual processing requirements for the reason that they need a target distribution in advance in the model training process.

In recent years, domain generalization (DG) has attracted much attention from scholars in DL-related fields. The aim is to train the model using existing source domain (SD) data and test it on target domain (TD) data with unknown distribution. This is highly consistent with the actual needs of many industries. Ren *et al.* [14] proposed domain-invariant feature fusion networks (DIFFN) fusing intra-domain and inter-domain invariant features to improve the model generalization under unknown target conditions. Ren *et al.* [15] proposed domain fuzzy generalization networks (DFGN) with domain fuzzy and metric learning strategies, enhancing the capabilities of learning domain-invariant and discriminative features. Recently, the emergence of attention-based DG has stimulated interest in new architectures. Attention mechanisms can achieve adaptive weighting of high-value spatial/channel/time/branch through dynamic adjustment and obtain more relevant information to the task [16]. A large number of scholars have carried out research on DG based on attention mechanisms. Ng *et al.* [17] proposed a channel-wise and spatial-wise hybrid domain attention mechanism that achieved domain generalization on several datasets. Hua *et al.* [18] captured the critical features of fundus images in channel, position, and spatial regions with a multi-scale attention mechanism. In fault diagnosis, attention mechanisms have also been proven to improve model generalization [1920].

In this work, as shown in Fig. 1, DG is divided into multi-source domain generalization (MSDG) and single-source domain generalization (SSDG) according to [21], that is, the number of domains involved during model training. Shen *et al.* [13] introduced a domain generalization network combing invariance and specificity, employing a knowledge extraction branch for domain invariant features via correlation alignment and triplet losses and a similarity indication branch for diagnostic decisions by selectively combining prediction scores based on similarities. Shi *et al.* [22] proposed a reliable feature-assisted contrastive generalization net that learns device-independent knowledge and enhances generalization with a feature-assisted module. Li *et al.* [23] applied domain enhancement, utilizing adversarial training for domain-invariant features. Li *et al.* [1] proposed a semi-supervised multi-source *meta*-domain generalization method for tool wear state prediction. However, challenges remain in applying MSDG to practical engineering. For instance, what level of DDD among multiple SDs is necessary to ensure the model's generalization? How many SDs are necessary for the model's generalization, and for which classes of TDs is it effective?

For SSDG, the prevailing technological emphasis lies in expanding the data distribution in the SD with adversarial domain enhancement [2425] or direct transformations of inputs. Wang *et al.* [26] proposed multi-scale style generative and adversarial contrastive networks (MSG-CAN) with a multi-scale style generation strategy to ensure that the generated samples contain abundant state information. Cugu *et al.* [27] argue that consistent interpretations should be provided when enhancing the SSDG. Under this concept, different visual corruptions are used to corrupt the original input randomly and combined with the attention mechanism, enhancing the model to pay attention to the same semantics through uniform loss of attention constraints to improve the model's generalization. Fan *et al.* [25] proposed an adaptive standardization and rescaling normalization. The impact on DG of the statistics of normalization layers is studied. Bayasi *et al.* [28] proposed an additional network, BoosterNet, which can be directly added to any network. With a unique feature culpability measure, BoosterNet is episodically trained on critical unit data features that contribute most and least to class-specific prediction errors within the core network. Ouyang *et al.* [29] focused on intensities and texture-acquisition shifts caused by different acquisition processes in medical images, taking a causal-inspired approach to data enhancement. In the industrial field, Zhao and Shen [30] proposed an adversarial mutual information-guided single-domain generalization network (AMINet). Specifically, AMINet evaluates and minimizes mutual information upper bound using contrastive log-ratio to generate samples with novel distributions. Oord *et al.* [31] assessed InfoNCE for mutual information lower bound, utilizing supervised contrastive loss to maximize it. Ultimately, they achieved single-domain generalization-based fault diagnosis.

Existing SSDG methods increase the possibility of overlapping between SDs and TDs. However, these approaches exhibit randomness, particularly in industrial signals such as tool wear monitoring. The challenge persists with slight degradation symptoms and strong interference, further increasing the uncertainty of effective distribution expansion. Hence, a pivotal challenge addressed in this paper is establishing the correlation between the generated domain (GD) and TD distribution while expanding data based on a single SD. As consistent with the ideas of [27], this paper holds that the consistent interpretation between SD and TD should be considered when data diversity enhancement is carried out. Therefore, this paper proposes a discontinuous physical property-constrained SSDG network (DP2Net) for milling tool wear monitoring. A spatial attention mechanism [32] is used first to constrain the discontinuous periodic physical properties in the signal. Then, diversity generation is carried out based on the features under this constraint. Besides, a standard trend sample based on the milling discontinuity mechanism is introduced to constrain the physical properties further. Finally, the attention-weighted and generated features are used to train together and improve the model's generalization.

The three main contributions of this article are summarized as follows.

1) The DDD in tool wear monitoring is defined, especially whether it exists under the same working condition, which is analyzed in detail. This helps to determine which machining conditions the model is suitable for a TD in advance. According to our literature search, this article is the first to define and address SSDG-based tool wear state monitoring.
2) A feature constraint and generation network that considers the consistent interpretation of features of both SD and TD called DP2Net is proposed. The discontinuous periodic physical properties in milling establish the correlation among SD, GD, and TD. This makes up for the lack of correlation between generated data distribution and TD based on SSDG. The generated features have the same physical properties as the TD.
3) The generalization ability of the DP2Net under different DDD is analyzed through experiments.

The rest of this article is organized as follows. Section 2 details the DDD problem in tool wear. Section 3 describes the proposed method in detail. Section 4 presents the experimental results and analyses. Finally, conclusions are drawn in Section 5.

2. Data distribution discrepancies in tool wear

In DL, the DDD refers to the metric discrepancy in the distribution fitted between the training and test data, also known as covariate shift [33]. In this paper, a practical application-oriented question is put forward: which factors will lead to DDD? The existing definition of DDD is based on statistics and is the measurement after the discrepancies are generated, cannot determine the cause of the discrepancies in advance. As a result, the trained model cannot be determined for which scenarios to use effectively.

The situation of DDD in the tools wear monitoring task is analyzed to clearly state the problem to be solved. At present, for tool wear monitoring, as shown in Eq. (1), whether the DDD exists mainly depends on whether the process elements (*PE*) such as equipment, tool structure/materials, and workpieces change; that is, there will be significant discrepancies in data distribution across working conditions (changes in any *PE*). This is also the main scenario in the current academic field when discussing model generalization.

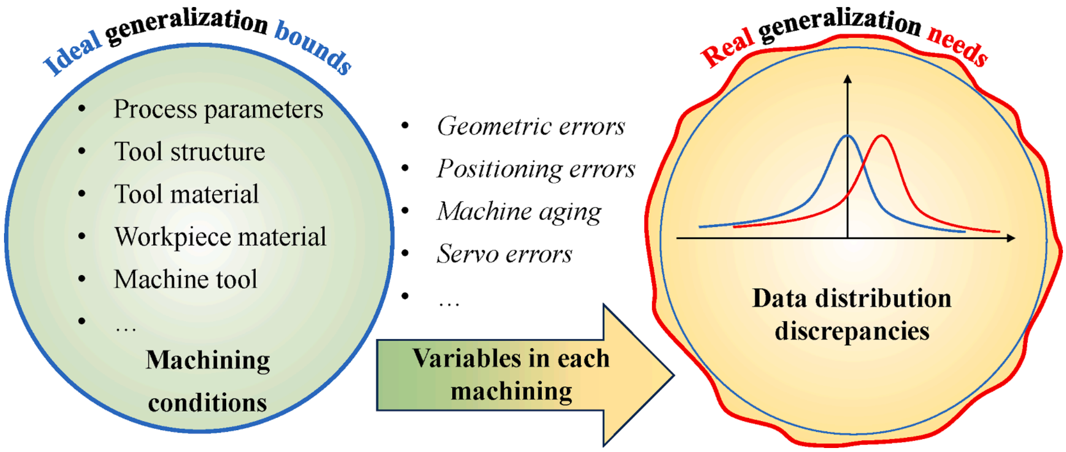

Fig. 2. DDD caused by process systems.

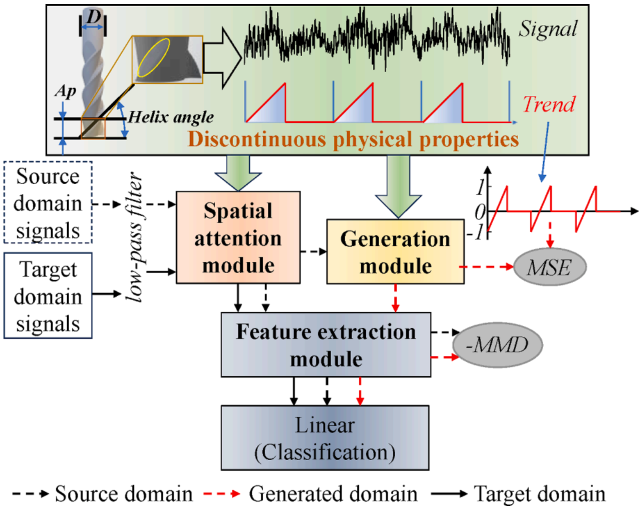

Fig. 3. Framework of proposed method.

$$\begin{cases} D = 0 & if: PE_i = PE_j \\ D = 1 & if: PE_i \neq PE_j \end{cases} \tag{1}$$

where D refers to whether the DDD exists. When it is 1, it means they exist, while 0 means there is no discrepancy. The subscripts i and j represent any process elements under different cutting tasks. PE is defined as follows:

$$PE = \{M, T, W, P_p\} \tag{2}$$

where M refers to the machine tool, T refers to the tool, W refers to the workpiece, and P_p refers to the specific process parameters.

The DDD is essentially caused by different incentives for the tool wear process. Most existing studies believe that the data obey the same distribution with the same equipment under the same working conditions [23]. That is when the PE of the processing task in Eq. (2) does not change, it is considered that there is no DDD. Tools are usually mass-produced and mass-used in actual processing, even with the same equipment and process parameters for cutting. In this case, the PE in Fig. 2 does not seem to be changed. However, the actual situation is that the performance of the M is always in a slow decay process due to the wear and impact of the mechanical parts [34]. At the same time, in each process, there are also W and T position errors [35]. In addition, even if the same type and batch of tools are being manufactured, there will be geometric errors, resulting in inconsistent tool edge lines, spiral angles, and other geometric parameters.

Therefore, in the actual processing scenario, PE is:

$$PE = \{M(t), T(g_t, p_t), W(g_w, p_w), P_p\} \tag{3}$$

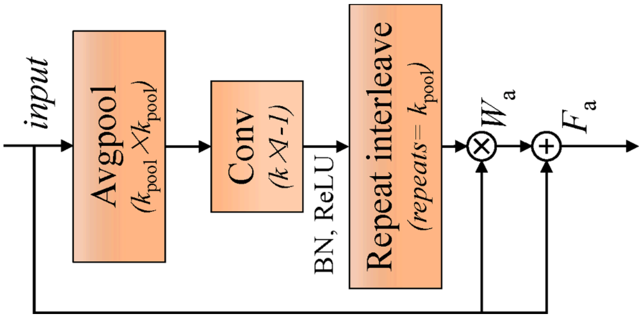

Fig. 4. Structure of S. BN demotes Batch Normalization Layer. ReLU denotes Rectified Linear Unit.

where $M(t)$ indicates that machine performance is a function of time. $T(g_t, p_t)$ represents the tool's geometric and position error. $W(g_w, p_w)$ represents the workpiece's geometric and position errors.

The above factors are real in practice, but there is a lack of relevant technology to measure them comprehensively and accurately in real-time. In short, the elements in Eq. (3) can never be consistent under different cutting tasks. In the tool wear monitoring task, the DDD always exists. This phenomenon can be demonstrated by the performance of existing tool wear monitoring models [836]. Data verification of this conclusion will be discussed in section 4.

To sum up, this paper aims to improve the generalization of the tool wear monitoring model, that is, the model trained with one processing (source domain) data can be directly applied to other processings (target domains).

3. Proposed methods

The proposed method framework is shown in Fig. 3. It contains a discontinuous physical property-guided spatial attention module S, a domain generation module G, and a feature extraction module F.

The proposed framework complies with the ideal of data-mechanism fusion. Specifically, the signal length corresponding to each tooth is calculated by sampling rate, sindle speed, and number of teeth. and the result is used as the design criterion of the convolutional kernel in module S. This is for extracting the signal fragment corresponding to each tooth (that is, related to wear) in the signal. A similar design is carried out for the convolution kernel in module G. In addition, a standard trend vector is constructed to constrain the GD according to the axial cutting depth in the process parameters, the tool structure parameters (diameter, helix angle, number of teeth), and the physical properties of the monotonically increasing wear trend.

3.1. Cross-domain consistency constraints: Discontinuous physical property-guided spatial attention module

As mentioned in the introduction, it is difficult to establish the correlation between the SD and the TD due to the lack of specific information on the TD. As a result, the GD has strong randomness and lack of interpretability. This work proposes a cross-domain consistency constraint method based on discontinuous physical properties to guide spatial attention [32]. Specifically, the method mainly calculates the data distribution formed by the contact between the milling tool flank and the workpiece through the tool structure parameters and process parameters. This result is taken as the size of the convolution kernel, which is calculated as follows:

$$k = \frac{\left(f_s * \frac{60}{N_{\text{speed}}}\right)/n}{k_{\text{pool}}} \tag{4}$$

where f_s is the sampling frequency of raw signals (Hz), N_{speed} is spindle speed (RPM), n is the number of teeth, and k_{pool} is the kernel size and stride set for the first average pooling layer.

For milling processing, each tooth is gradually in contact with the workpiece. That is, the signal obtained by the tool rotation of one circle can be divided into n segments according to the number of teeth (n), and each segment also contains the corresponding signal of the groove (not flank). Eq. (4) calculates the corresponding signal segment of each tooth after downsampling. The result of Eq. (4) is taken as the receptive field of the convolution layer in S. So that S receives the signal length corresponding to one tooth each time and adaptively captures the signal corresponding to the flank.

The structure of S is shown in Fig. 4.

With S, all signals will be constrained by discontinuous physical properties at the same spatial scale, which can obtain the "pure" signal fragments that are theoretically most relevant to tool wear at the spatial scale. In theory, features (F_a) weighted by attention weight (W_a) are more semantically efficient than the raw signal, so it will be used as the input of G.

3.2. Generation and constraints

The structure of G is shown in Fig. 5, which includes one pooling layer, three convolution layers, and a transposed convolution

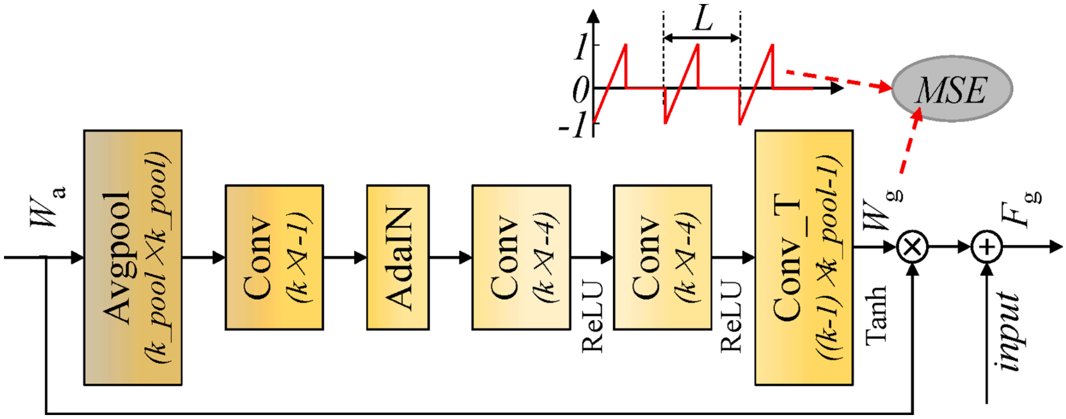

Fig. 5. Structure of G.

layer. Among them, except for the number of channels of the convolutional layer (1, 4, and 4, respectively), other parameters of the pooling and convolutional layers are consistent with those in S. More importantly, AdaIN [37] further increases the diversity in the generation process.

This module introduces a standard trend vector (V_{st}) derived from discontinuous physical properties, which is used to constrain the basic trend of generated samples to establish the association with other domains. V_{st} is a series of numbers of equal length to input, which is periodically composed of monotonically rising intervals from -1 to 1 and continuous 0 values, as shown in Fig. 5. Specifically, the percentage P of the length of each monotone rising interval in the total length of the series is:

$$P = \frac{Ap/\tan\beta}{(D \times \pi)/n} \tag{5}$$

where Ap is the axial cutting depth (mm). β and D are the helix angle (°) and the tool's diameter (mm).

The total length L of the monotone interval and the 0-value interval is:

$$L = \frac{f_s \times (60/N_{\text{speed}})}{n} \tag{6}$$

With MSELoss, V_{st} and the generated weight (W_g) are constrained so that W_g is implanted with discontinuous physical properties consistent with other domains. The raw input is then weighted to give the generated features (F_g).

$$L_{\text{MSE}} = \frac{1}{n_S}\sum_{i=1}^{n_S}\left(W_g^i - V_{st}\right)^2 \tag{7}$$

where n_S denotes the amount of SD's sample.

3.3. Training and optimization

This paper uses deep convolutional neural networks with wide first-layer kernels (WDCNN) as F. As a classic network model, WDCNN has a good suppression effect on the noise in physical signals and has a wide range of applications in industry, so it is often used as a basic network. The structure and parameters adopted are consistent with those in [38]. The probabilities of different classes are output by a linear layer.

The training of the proposed method consists of two steps. Specifically, S and F are first trained on a single SD to obtain optimized S. The trained S is used for spatial weighting of the SD, the weighted features are input to G for diversity generation, and the weighted features are input to F for generalization training.

In the first step, S and F are optimized with minimal cross-entropy loss:

$$L_{\text{CE}} = -\sum_c y_S^c \log Y_S^c \tag{8}$$

where y_S, Y_S are real and predicted labels, respectively. The superscript c denotes different categories.

The second step measures the distance between the SD and GD features before the linear layer by maximum mean discrepancy (MMD) [39]. The MMD maximization is used for optimizing G, to maximize the diversity of the generated features.

$$L_{\text{MMD}} = \sqrt{\frac{1}{n_s^2}\sum_{i,j=1}^{n_s} k\left(x_S^i, x_S^j\right) - \frac{2}{n_s^2}\sum_{i,j=1}^{n_s} k\left(x_S^i, x_G^j\right) + \frac{1}{n_s^2}\sum_{i,j=1}^{n_s} k\left(x_G^i, x_G^j\right)} \tag{9}$$

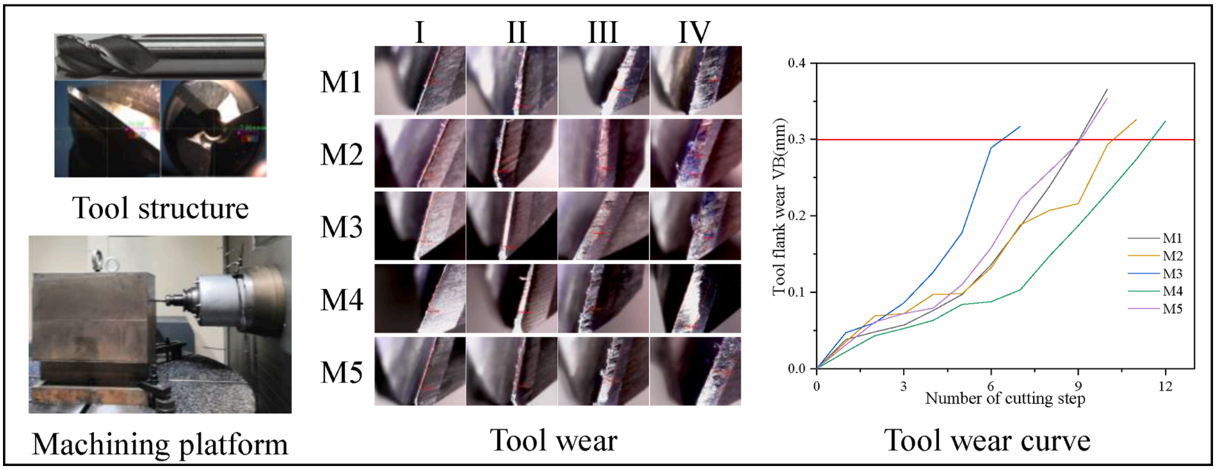

Fig. 6. Machining experiment process.

where $k(x_S, x_G) = \exp(-\|x_S - x_G\|^2/2\gamma^2)$, γ is the width of the nucleus. x_S, x_G are TD and GD samples, respectively.

Then, G is optimized by:

$$L_G = L_{MSE} - \alpha^* L_{MMD} \tag{10}$$

where α is a weight coefficient set as 20 in this paper.

Finally, the task loss is minimized to enable F to be generalized to diverse distributions:

$$L_{task} = \frac{1}{n_s}\sum_{i=1}^{n_s} L_{CE}(Y_S^i, y_S^i) + \frac{1}{n_s}\sum_{i=1}^{n_s} L_{CE}(Y_G^i, y_G^i) \tag{11}$$

In summary, the proposed method is shown in Algorithm 1.

Algorithm 1:

Pre-training stage Input: Source dataset $Data_S = \{x_S^i, y_S^i\}_{i=1}^{n_S}$ with Fourier low-pass filtering. Model: S and F. for $epoch = 1$ to 100, do Randomly sample (x_S^i, y_S^i) from $Data_S$. Train S and F using Eq. (9). Return: trained S. # **Training stage** Input: Source dataset with Fourier low-pass filtering. Model: *G, F, and* trained S. for $epoch = 1$ to 100, do Randomly sample (x_S^i, y_S^i) from $Data_S$. Train G using Eq. (10). Train F using Eq. (11). Return: Trained F. # **Inference stage** Input: Target datasets with Fourier low-pass filtering. Model: Trained S and F.Output: Final decisions.

Besides, the monitoring signal is the coupling of the entire process system. Especially, the interference of the machine tool components, and it is clear that such interference will affect the decision-making preferences of the model. Generally, the performance of the machine tool is relatively stable in the short term, so it can be regarded as a natural frequency. Besides, the frequency related to tool wear is mainly concentrated in the low-frequency part. Therefore, this paper first carries out low-pass filtering for signals.

4. Experimental study

4.1. Dataset description

1) PHM2010[40]. The dataset came from the PHM 2010 challenge. Three ball-nose carbide milling tools with three flutes were tested in the same machining conditions. Tool life cycle signals were collected and sampled at 5 kHz.
2) Machining experiments. The same type of three-tooth integral end mill (material: carbide, diameter: 16 mm, helix angle: 35°) was used in the experiment. The experiment is tested on a 5-axis CNC milling center THM6380 IV, and the workpieces are carbon steel 40. Specifically, 420 mm was continuously milled each time in a vertical straight line along the y direction, and then the width of flank wear was measured in situ. The experimental process is shown in Fig. 6.

All data are divided into three categories according to different degrees of P_p in Eq. (3). To test the effectiveness of the proposed method under different working conditions (different distribution differences). Specifically, case I is based on Eq. (2), which is used to verify that there are DDD in the same working conditions. Any process parameter variation is set in case II to amplify DDD. In case III, all process parameters except the spindle speed are completely changed. The specific parameter settings are shown in Table 1.

Where N is the spindle speed, RPM. Ap is the axial cutting depth, mm. Ae is the radial cutting depth, mm. f is the feed rate, mm/min. C1, C4, and C6 are from the PHM 2010 dataset; other tools are from machining experiments.

Table 1
Eight machining conditions' parameter settings for varying data distribution discrepancies.

	Tools	P_p			
		N	Ap	Ae	f
Case I	C1, C4, C6	10,400	0.2	0.125	1555
	M1, M2	1200	3	1.8	180
Case II	M3	1200	3	2	180
	M4	1200	3	1.8	200
Case III	M5	1200	4	2	220

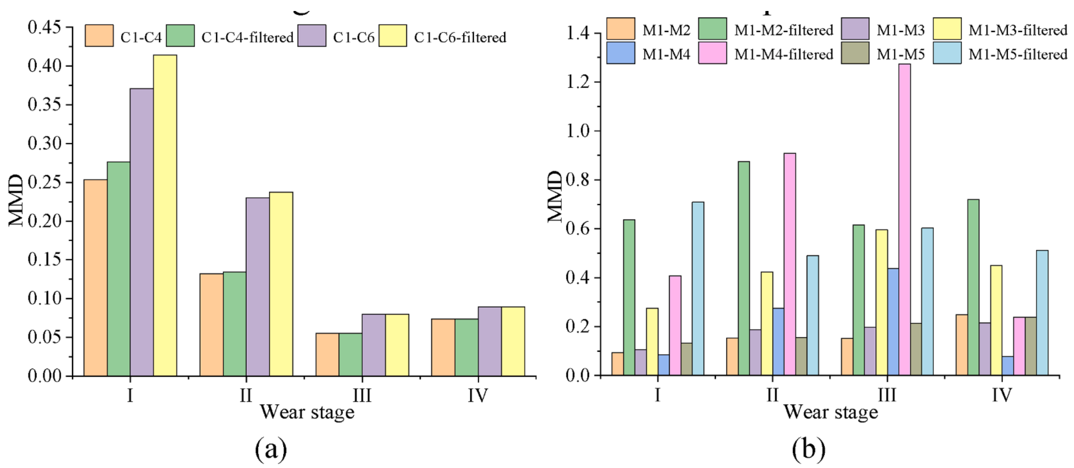

Fig. 7. DDD measurement results of (a) PHM2010 data and (b) machining experiments data.

4.2. Data distribution discrepancies measurement and preprocessing

First, MMD is used to measure the DDD between signals under different working conditions. The result is shown in Fig. 7(a). The data of PHM 2010 belongs to the constant condition, but the DDD among C1, C4, and C6 is obviously large. As shown in Fig. 7(b), it is the DDD measurement result of machining experiment data. It can be found that the DDD under different process parameters and wear stages are inconsistent, which is consistent with Eq. (3) defined in this paper. At the same time, the uncertainty of the influence of working conditions on the DDD is also proved.

As mentioned above, the vibration signal is low-pass filtered. Specifically, the PHM 2010 dataset's low pass frequency is set at 1733 Hz (spindle frequency less than six times [7]), while the low pass frequency of the machining experiments data is set at 600 Hz (10 times of the tooth pass frequency [32]). By comparing the DDD between the original signal and the signal processed by low-pass filtering, it can be found that the homologous noise in the original signal will reduce the DDD of different working conditions. This results in much noise as features while ignoring the high-value signals associated with tool wear, resulting in a fake generalization improvement. Therefore, it is necessary to filter the signal.

Then, all signals are separated into initial wear (I), steady wear (II), and accelerated wear (III) according to [41] and failure stage (IV, while mean VB $>$ 0.3 mm, GB/T 16460–2016). For the PHM 2010 dataset, the force signal in the x-direction is taken as the monitoring data. The length of each sample is 4608 (contains 16 cycles of data). According to the method in [36], each tool in the PHM2010 data set is divided into different states, too. Each class contains 2000 samples, respectively. C1 is SD, and C4 and C6 are TDs. For the machining experiments dataset, the length of each sample is 6000 (contains 12 cycles data), in which M1 and M2 each class contains 250, 400, M3, M4, and M5 each class contains 250, 400, and 400 samples, respectively. Besides, the vibration signal in the x-direction is taken as this experiment's monitoring data to verify the proposed method's effectiveness on different signals and consider the cost and convenience of obtaining signals in actual processing. M1 is the SD. The data of other tools is used as TDs. Besides, due to the introduction of spatial attention, each segment aligned in time was performed [7]. Finally, SD is divided into two parts, with 70 % as the training set and 30 % as validation set. All TDs are used as test sets.

4.3. Comparative methods and results

To verify the validity and superiority of the proposed method, several successful and related DG methods are used for comparison. Besides, several ablation experiments are implemented. All methods are carried out on the i7-11800H 16 GB platform with Python 3.9.7-Pytorch 1.10.1 and NVIDIA's GeForce GTC 3060 GPU. The code implementation of the methodology involved in this study is as follows:

Table 2
Experimental results of all methods in different cases (accuracy %).

		Case I			Case II		Case III
Method	Task (TD)	1 (C4)	2 (C6)	3 (M2)	4 (M3)	5 (M4)	6 (M5)
(1)	PDEN	76.13	86.93	75.98	86.28	67.17	76.52
(2)	AMINet	63.74	87.45	77.54	72.96	78.73	81.01
(3)	WDCNN(raw)	82.21	71.59	58.01	61.98	86.13	65.67
(4)	WDCNN(filtered)	78.70	69.39	62.22	78.15	77.49	62.29
(5)	*S*&WDCNN(raw)	74.76	55.71	66.05	78.93	79.76	66.54
(6)	*S*&WDCNN(filtered)	80.17	76.86	65.20	90.55	84.83	76.63
(7)	*G*&WDCNN(filtered)	**90.98**	76.00	83.04	85.27	84.32	84.50
(8)	**Proposed**	90.91	**87.66**	**92.05**	**93.01**	**91.64**	**88.57**

* Tasks 1 and 2 use C1 as SD. All other tasks use M1 as SD.

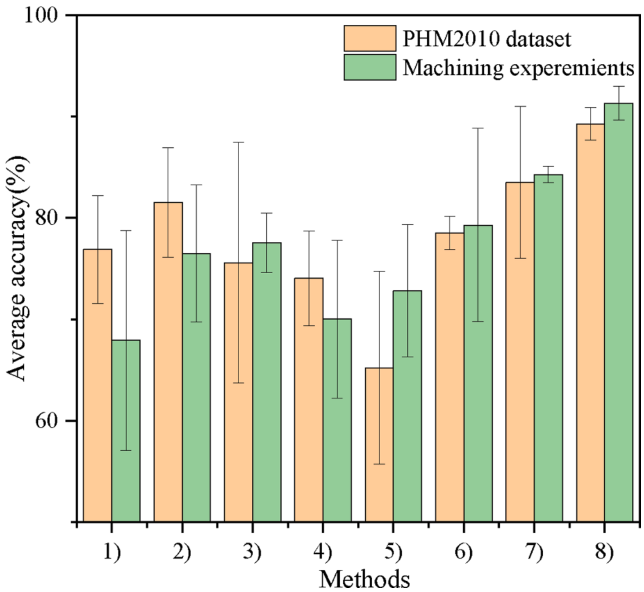

Fig. 8. Average accuracy and standard deviation.

1) PDEN [42]. The progressive domain expansion network (PDEN) is a state-of-the-art SSDG method that continuously improves the diversity of generated data by gradually training new generative models and imposing consistency constraints on each round of the generative model training process. The PDEN code can be obtained at https://github.com/lileicv/PDEN.
2) AMINet [30]. As mentioned in the introduction, AMINet is the only method for SSDG fault diagnosis for now. Samples are generated based on the features after the Fourier transform as input, and mutual information is used to restrict the upper and lower bounds of the generation process to improve the generalization of the model in other fields.
3) WDCNN [38] (with raw signal). WDCNN is a classic fault diagnosis method. It is often used as a basic method of fault diagnosis.
4) WDCNN (with Fourier low-pass filtered signal). Unlike the previous method, which directly uses the raw signal as input, this method uses the signal after low-pass filtering as input.
5) *S*&WDCNN (with raw signal). The proposed *S* combined with WDCNN takes the raw signal as input without *G*.
6) *S*&WDCNN (with Fourier low-pass filtered signal). Similarly, the difference with the previous method is that the input signal is processed after low-pass filtering.
7) *G*&WDCNN. The proposed *G* combined with WDCNN takes the filtered signal as input without *S*.
8) The proposed method.

For PDEN and AMINet, the parameter Settings in the original work are adopted. The batch size of all methods is set to 64. For methods 3)-8), the learning rate of the Adam optimizer was set as 0.001. The k_{pool} is the undersampling of the pooling layer, so it is set to 4 for high computational efficiency. The α was set as 20 (experimentally tried).. The k of the PHM 2010 dataset was estimated to be 25 and the k of the machining experiments dataset was calculated to be 41. The learning rate's cosine attenuation strategy was applied to the training process of the feature extraction module, and the attenuation period was 20 epochs. Moreover, the learning rate for G was always 0.00001, and the Adam optimizer was also adopted.

The experimental results of all methods are shown in Table 2.

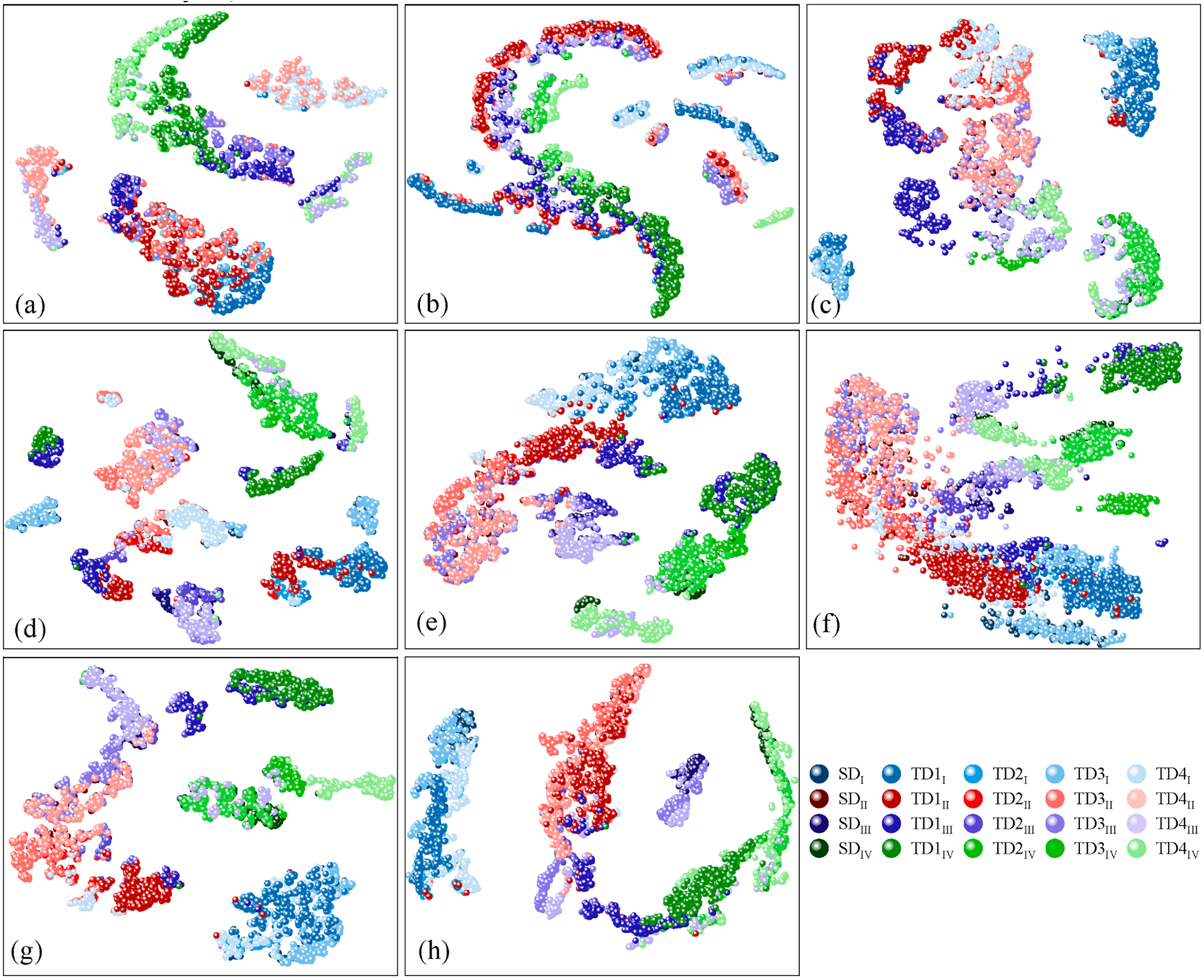

Fig. 9. T-SNE visualization. (a)-(h) correspond to methods 1)-8) involved in this study, respectively.

In addition, the average accuracy and standard deviation of different methods on the two datasets were counted, and the results are shown in Fig. 8. The proposed method achieves the best results in terms of average identification accuracy on both datasets. Meanwhile, the standard deviations are 1.63 on the PHM 2010 dataset and 1.66 in the machining experiments.

4.4. Analysis

4.4.1. Results analysis

T-SNE visualizes the features extracted under different methods (before the classification layer).

From Fig. 9(a), PDEN achieves separation among different classes, but classes I and III overlap with classes II and IV, respectively. In Fig. 9(b), for AMINet, the aliasing among different classes is not serious, but the distance is relatively close, which is not conducive to the discrimination of different classes. Compared with Fig. 9(c) and (d), it is conducive to improving the distance among classes when the signal is filtered. Still, it does not promote the aggregation of similar features. From Fig. 9(e) and (f), with the addition of S, the intra-class aggregation is improved, but the distance among classes is not improved well. At the same time, it can be found that in Fig. 9 (f), compared with the previous method, when S is applied to the filtered signal, the features become more divergent compared with the raw signal. On the one hand, the large number of homologous noises mentioned in Fig. 4 will bury the weak wear difference. On the other hand, adding S restrains the inter-domain consistency. However, the F lacks corresponding constraints, so the feature is more divergent. From Fig. 9(g), adding G constraints improves intra-class and inter-class distances. However, due to the lack of consistency and constraint of S on SD and TD, the constraint of the inter-domain distance of similar domains is poor. Fig. 9(h) shows better inter-class distance and intra-domain aggregation.

Moreover, a set of confusion matrices of the proposed method is created under each task, as shown in Fig. 10. For case I (tasks 1–3), misjudgment mainly occurs between classes III and IV. For tasks under the same working conditions, accurately identifying the failure stage and the accelerated wear is always the most challenging task. Class III is misjudged as II for case II (task 4, task 5). For case III

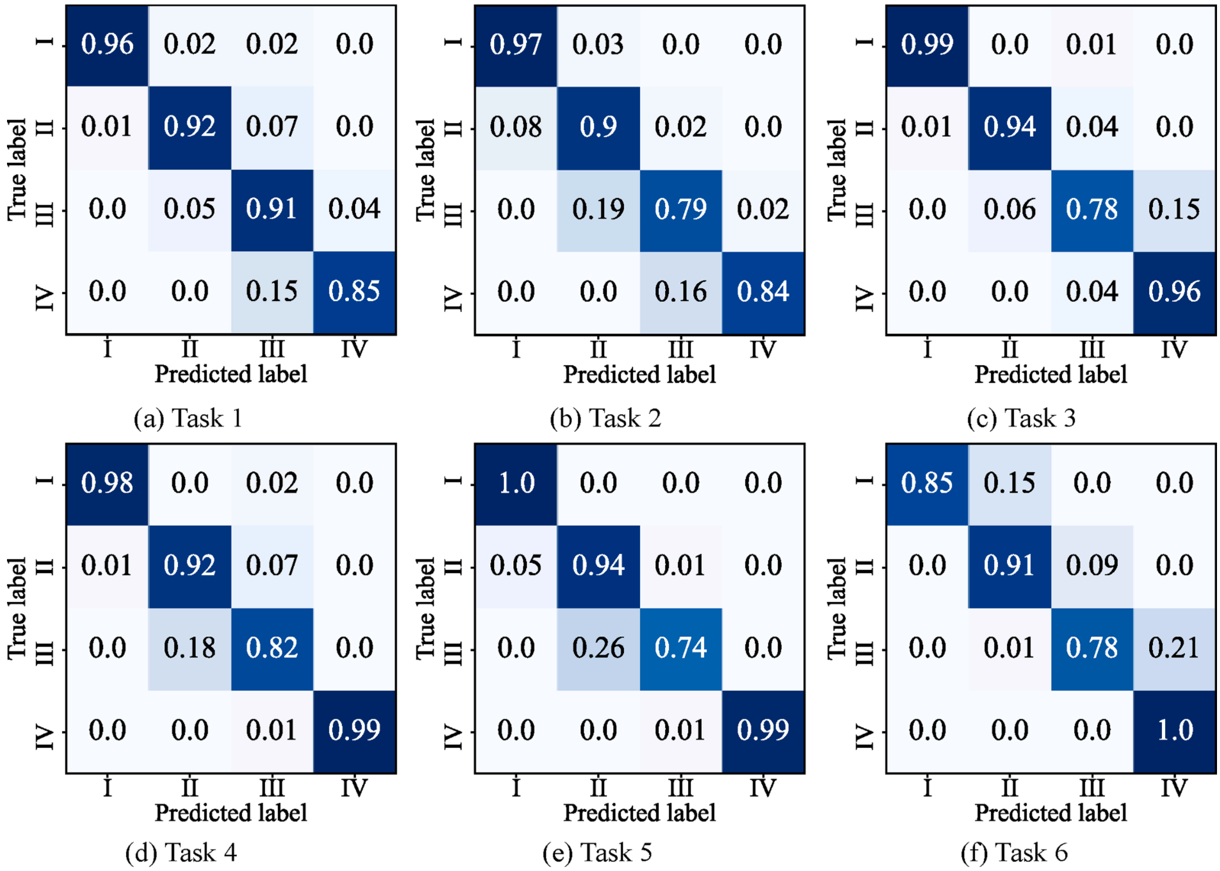

Fig. 10. Confusion matrix visualization. (a)-(f) correspond to tasks 1–6 involved in this study, respectively.

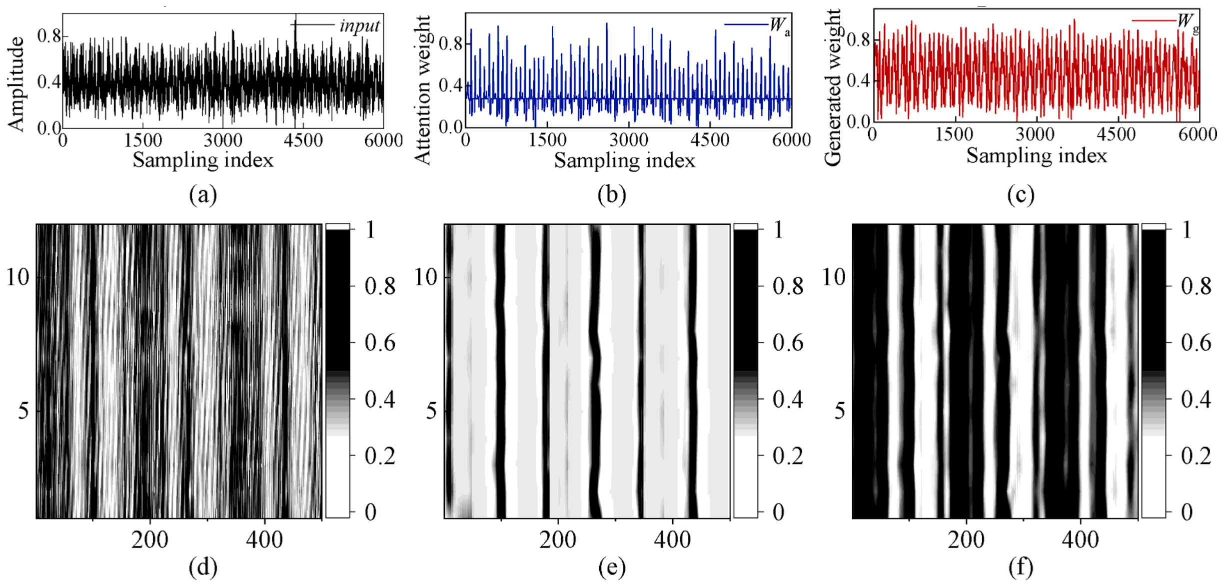

Fig. 11. Consistency between .
Source and generation domains. (a)-(c) are the waveforms after standardization of input, W_a, and W_g, respectively. (d)-(e) are their heat map

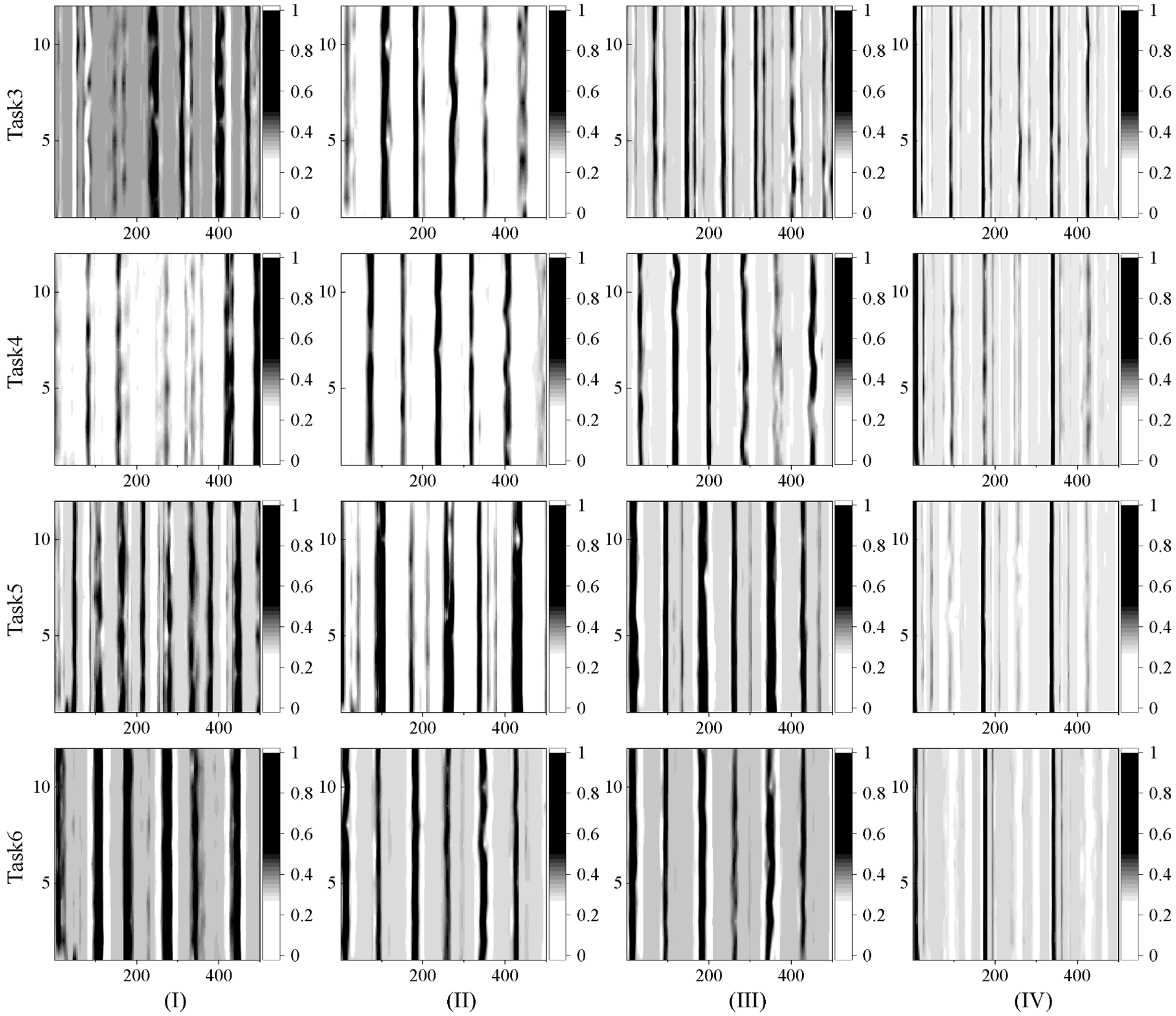

Fig. 12. Consistency of target domains. The hot maps from top to bottom are tasks 3–6, and from left to right are wear stage I-IV.

(task 6), the obvious trend is that the wrong prediction results are towards the more severe wear stage. From the working condition setting of case III, it is easy to think that the cutting amount of the tool is much larger than SD. That is, compared with the tool in SD. The tool in this TD will produce greater cutting resistance even at the same wear stage. One of the characteristics of severe tool wear is that the resistance will increase so that the model may give a more severe assessment.

4.4.2. Consistency analysis

Here, the consistency among SD, GD, and TD is visually analyzed. Firstly, in Fig. 11(a)–(c), the input sample *input* (signal after low-pass filtering) and its corresponding W_a, and W_g are shown, respectively. Specifically, for better comparison, all the data/features were normalized. Further, the data point (500) corresponding to each rotation of the tool is reshaped into [500 × 12] and then plotted into the heat map form as shown in Fig. 11(d)–(f).

By comparing Fig. 11(d) and (e), it can be found that the W_a obtained after the *input* is weighted by S shows a strong periodic discontinuity that conforms to the theoretical expectation (black vertical stripe in Fig. 11(e)), which means that S pays special attention to a part of the input. More importantly, this discreteness maintains a high consistency in every turn of the tool rotation. As shown in Fig. 11(f), the diversity of the generated feature W_g is enhanced in the region emphasized by W_a, but it still maintains a good periodic discontinuity as a whole. Therefore, it can be considered that a consistency between the SD and the GD satisfies the theoretical expectation.

The same visualizations are carried out for all TD's data. As shown in Fig. 12. All the different stages of TDs show the above periodic discontinuity.

Further, it can be found that for all tasks, as the wear increases, the area of focus of S becomes more concentrated. On the one hand, the wear form in the early stage is mainly run-in wear, which does not form a strong incentive signal. So, it is difficult to effectively distinguish the cutting signal from the non-cutting signal segment. On the other hand, as the tool wear increases, the incentive formed by the flank at the moment of contact with the workpiece is stronger, thus forming a stronger cycle boundary. In addition, there are some differences in the performance of S in different TDs, mainly in the degree of suppression of unimportant regions. However, this does not affect the periodic discontinuity and final accuracy.

To sum up, it is reasonable to believe that the method proposed in this paper establishes an interpretable consistency constraint among SD, GD, and TDs that meets the theoretical expectation.

5. Conclusion

This paper details and defines the data distribution discrepancies in tool wear monitoring. Unlike the existing postmortem statistics, this paper's definition is an ex-ante one, which helps determine the applicable condition of the monitoring model. A discontinuous physical property-constrained single-source domain generalization network is proposed to address the cross-conditions milling tool wear state recognition. A reliable consistency association is established among source, target, and generating domains. An open dataset and extensive machining experiments verify the generalization of the proposed method under different distribution discrepancy levels. The proposed method achieves a 12.69 % improvement in average accuracy on the PHM 2010 dataset and a 23.37 % improvement in the machining experiments, compared with the benchmark results of WDCNN. Feature visualizations provide consistency and interpretability.

In future work, many factors should be considered, such as the impact of different spindle speeds on the spatial distribution of signals. Spindle speed is a crucial and variable process parameter in actual machining. When the spindle speed changes, the signal length corresponding to one tool rotation will change; the signal will be deformed in space (low speed means elongation, high speed means compression). Conventional networks, such as convolutional networks, have difficulty dealing effectively with the problem of signal spatial transformation, and it is difficult to ensure spatial invariance. Therefore, the consistency among source, generating, and target domains under different spindle speeds is worth exploring. Moreover, the generalization boundary of the model, that is, to what extent the generalization of a DL model is reliable under DDD, needs to be further studied.

CRediT authorship contribution statement

Xuwei Lai: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Kai Zhang:** Writing – review & editing. **Qing Zheng:** Writing – review & editing. **Minghang Zhao:** Writing – review & editing. **Guofu Ding:** Project administration, Conceptualization. **Baoping Tang:** Supervision, Conceptualization. **Zisheng Li:** Visualization, Validation, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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